

Data Description

The dataset contains sales transactions collected from **multiple warehouses**.

Each warehouse operates independently, which means:

- The same **region, city, sales representative, customer (pharmacy), or product** may appear under different names across warehouses.
- There is no shared naming convention between warehouses.
- The data contains duplicates, spelling variations, abbreviations, and otherwise inconsistent representations.

Goal:

Analyze the sales data intelligently despite these inconsistencies and the absence of standardization.

Columns

Column	Description
id	Unique row identifier for each transaction.
supplier_id	Identifier of the supplier that provides the product.
invoice_id	Invoice number associated with the transaction.
prod_id	Product ID in the system.
quantity	Number of units sold in the transaction.
product_name	Name of the product; may differ across warehouses for the same product.
discount	Discount applied to the transaction (percentage or value).
total_amount	Total transaction value after discount.
account_id	Unique customer/account identifier (a pharmacy).
account_name	Customer (pharmacy) name; may appear in different forms across warehouses.
account_address	Customer (pharmacy) address.
city	City where the transaction occurred; naming may be inconsistent.
region	Geographic region; often inconsistent across warehouses.
area	Sub-region or district; may be missing or inconsistently filled.

Column	Description
area_id	Identifier for the sub-region, when available.
user_id	ID of the user who recorded the transaction.
user_name	Username of the system user.
employee_name	Sales representative name; may appear in multiple formats for the same person.
creation_date	Timestamp of when the transaction was recorded.

Task

1. Data Cleaning & Unification

The data is inconsistent across warehouses. The same entities may appear with different names, including:

- cities
- regions
- sales representatives
- customers (pharmacies)
- products

Expected approach:

Build a unified entity-mapping strategy. Acceptable techniques include:

- fuzzy matching
 - NLP techniques
 - embeddings / AI-based tools
 - rule-based logic where appropriate
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2. Pharmacy-Level Invoice & Revenue Analysis

Once the pharmacy names have been mapped and the data unified, work at the pharmacy level.

Expected work:

- Filter / group the invoices by each pharmacy.
- Compute how much revenue each pharmacy generated.

- Break down each pharmacy's revenue by the supplier it purchased from.

The result should make it possible to answer, for any pharmacy: which invoices belong to it, how much revenue it represents, and which suppliers that revenue came from.

3. Data Linking & Modeling

There are no global keys across warehouses. Think in terms of **entity resolution**.

Expected data model:

- Fact table: sales transactions
 - Dimensions:
 - products
 - customers (pharmacies)
 - sales representatives
 - geography (city / region / area)
 - suppliers / warehouses
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4. Analytics

First analyze each warehouse separately, then analyze the unified data after cleaning.

Expected insights:

- top products
 - top customers
 - top sales representatives
 - best-performing regions
 - detection of data inconsistencies
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5. Machine Learning

- Sales forecasting
- Demand prediction
- Anomaly detection, covering:
 - unusual sales patterns
 - inconsistent representative performance
 - warehouse-level anomalies

Note:

AI / AutoML is allowed, but **understanding the data problem is more important than the model itself.**

6. Dashboard / Visualization

Build a clear dashboard showing:

- sales by region
- sales by representative
- sales trends over time

Compare the view **before** and **after** data unification.

7. Automation (Telegram Reporting)

Send periodic reports via Telegram, including:

- total sales
 - top region
 - top sales representative
 - alerts for drops / anomalies
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